Information Retrieval

Practical session n°4: Evaluation

Create a directory named *practice4*. In this directory, create a new file named *practice4_report.txt*.
During the practical session, for each exercise, copy-paste some outputs of your program into this file to demonstrate that you have completed the exercise and it works correctly. Add some explanations.
At the end of the practical session, copy-paste the source code of your program(s) in the directory *practice4*.
Compress the directory in a file named *practice4_YourTeamName.zip* (e.g.: *practice4_VictorAlbertJules.zip*).
Upload this compressed file (one file / team) on the website of the course (**deadline November 30th** (first try); **December 7th** (final version)).

INEX runs

You will build several runs (maximum = 50 runs) for INEX "Ad-Hoc Relevant in Context" task:
- Each run will be stored in a text file, containing the results returned by your system for the 7 queries listed below. Each file must not exceed 10,500 lines.
- the filename of your runs should be named using the following template:
  TeamName_Run-Id_WeigthingFunction_Granularity_Stop_Stem_Parameters.txt

With:
- Run-Id = a unique identifier for each run
- WeightingFunction = the weighting function applied, such as ltn, ltc, bm25, etc.
- Granularity ϵ { articles, elements, passages }, i.e. the document unit. If "elements", you may specify the list of XML tags ϵ { article, header, title, bdy, sec, p, etc.} used as document units.
- Stop ϵ { nostop, stopN }, where N denotes the size of the stop-list.
- Stem ϵ { nostem, porter, lovins, paice, etc. }, specifying the stemming algorithm applied.
- Parameters: list any interesting additional parameters used, along with their value.

Example: VictorAlbertJules_12_bm25_elements_sec_p_stop344_nostem_k1.2_b0.75.txt

Please upload your runs (compressed zip files containing one or several runs, without any directory) on http://ri.gery.fr: General login: m2ri_student, password: df4kvjZGoSu4yPd / Individual login: your UJM email, password: sent by email.

Exercise 1: SMART *ltn* first run

Using your IR System, index the collection of the Practical session n°3 (tokenization, dictionary, *df*, *tf*, w_{td}):
- Simple tokenization: terms without digits or special characters.
- No stop-words list is used.
- No stemming is applied.
- SMART *ltn* weighting function.

Compute the following statistics:
- Total indexing + weighting time (#sec).
- Total number of tokens occurrences in the entire collection (#tokens), before any further processing (normalization, case folding, stop-words, stemming, etc.).
- Total number of distinct tokens in the entire collection (#distinct tokens), before any further processing.
- Average length of distinct tokens (#characters).
- Total number of terms occurrences in the entire collection (#terms).
- Total number of distinct terms in the entire collection, i.e. vocabulary size (#distinct terms).
- Average document length (#terms).
- Average length of vocabulary terms (#characters).
- Weight of the term "ranking" in document #23724.
- RSV (Retrieval Status Value) of document #23724, for the query « web ranking scoring algorithm ».
- Top-10: a list of the ten most relevant documents along with their RSV, for the query « web ranking scoring algorithm ».

Retrieve a ranked list of 1 500 articles, for each of these 7 queries:

Query id	Query
2009011	olive oil health benefit
2009036	notting hill film actors
2009067	probabilistic models in information retrieval
2009073	web link network analysis
2009074	web ranking scoring algorithm
2009078	supervised machine learning algorithm
2009085	operating system mutual exclusion

Using these results, build a run for the INEX RIC task. Ensure the syntax is correct (cf. lecture n°4 and the example file *ExempleRun_LTN_articles.txt*). Test your run on http://ri.gery.fr

Exercise 2: SMART *ltc* first run

Answer the same questions as in Exercise 1, but using the SMART *ltc* weighting function.

Exercise 3: *BM25* first run

Answer the same questions as in Exercise 1, but using the *BM25* weighting function. Set the k_1 and b parameters with these usual values: $k_1 = 1.2$ and $b = 0.75$.

Exercise 4: test runs

Build 12 runs (including the 3 runs of exercises 1-2-3), using stop-words (stop-words-english4.txt) or not, using a Porter stemmer or not: 3 (ltn|ltc|BM25) * 2 (stop-list) * 2 (stemmer) = 12 runs.
Include “test” in the filename of your runs:

TeamName_Run-Id_test_{ltn|ltc|bm25}_article_{nostop|stop671}_{nostem|porter}_Parameters.txt

Example: VictorAlbertJules_12_test_bm25_article_stop671_nostem_k1.2_b0.75.txt

Exercise 5: Tokenization, stemmer, stop-words, weighting

Build several runs using your chosen tokenization method, stemming algorithm, stop-words list, weighting function and weighting parameters.

Exercise 6: BM25 tuning

To optimize the BM25 weighting function, build runs that explore the two-dimensional space of its parameters (k_1 and b). Consider your optimization strategy carefully. A straightforward approach would be to set k_1 to 1.2 and test 11 values for b (ranging from 0.0 to 1.0, step = 0.1). Then, set b to 0.75 and test 21 values for k_1 (ranging from 0 to 4, step = 0.2). For a more efficient strategy, consider implementing the gradient descent algorithm.